Week 7

Example $f(x) = \frac{4x^3}{x^2+1}$

Show that f is invertable

$(f^{-1})'(2)$

f is invertable $\Leftrightarrow$ f is one-to-one

if $f(x_1) = f(x_2) \Rightarrow x_1 = x_2$

$$f'(x) = \frac{(x^2+1)\cdot 12x^2 - 4x^3 \cdot 2x}{(x^2+1)^2} = \frac{12x^4 + 12x^2 - 8x^4}{(x^2+1)^2} = \frac{4x^4 + 12x^2}{(x^2+1)^2} \geq 0$$